CMPSC 463 - Project 2
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4/26/2026

1. Project Description

This project develops an Evacuation Route Optimizer that models emergency evacuation scenarios in a conflict or disaster setting. The system represents a region as a weighted graph, where nodes correspond to locations and edges represent roads with varying levels of accessibility and safety.

The objective is to determine an optimal evacuation strategy that begins at a user-selected starting location, collects all refugees from designated nodes, and delivers them to one of several available refugee camps. The system operates under dynamic constraints, including damaged infrastructure and competing objectives of speed and safety.

A Streamlit-based interface allows users to interactively modify road conditions and execute the optimization process, providing immediate visual feedback of computed evacuation routes.

2. Project Significance

Efficient evacuation planning is a critical problem in humanitarian response, particularly in environments where infrastructure is damaged and information is incomplete. This project demonstrates how algorithmic techniques can be applied to support decision-making under such constraints.

The system is significant because it combines multiple computational challenges: graph-based shortest path routing, combinatorial optimization, and multi-objective cost balancing. It also models realistic trade-offs between travel time and safety, which are central concerns in real-world evacuation planning.

The project further illustrates how heuristic approaches can be used to obtain high-quality solutions in situations where exact optimization methods are computationally infeasible.

3. Code Structure Explanation

The project is organized into four main modules: graph generation, route optimization, visualization, and the Streamlit interface.

The graph generation module constructs a synthetic city-like network with clustered regions, random long-range connections, and designated node types for normal locations, refugees, and camps.

The route optimization module computes shortest path distances using Dijkstra's algorithm and applies heuristic methods to determine an efficient refugee pickup order and final destination.

The visualization module renders the network using PyVis, displaying nodes, edge conditions, and the computed evacuation route.

The Streamlit interface integrates all components, allowing users to configure scenarios, adjust parameters, and execute the optimization pipeline interactively.

4. Algorithm Descriptions

The evacuation network is modeled as a weighted graph where nodes represent locations and edges represent roads. Each edge has a base travel time and a risk value, which are dynamically modified based on road condition (intact, light, heavy, blocked). A parameter *alpha* controls the trade-off between speed and safety.

Shortest paths between nodes are computed using Dijkstra's algorithm. This ensures optimal routing between any two locations under the current edge weights. These results are precomputed into a distance matrix for efficiency.

The refugee ordering problem is treated as a Traveling Salesman Problem variant. Since exact solutions are computationally expensive, a greedy nearest-neighbor heuristic constructs an initial route.

This initial solution is improved using a 2-opt local search algorithm, which iteratively swaps route segments when improvements in total cost are possible. The final step selects the best camp based on total route cost from the last node.

5. Algorithm Verification

Dijkstra's algorithm was verified using small toy graphs with 3–5 nodes. In a toy graph, a start node had two possible paths to a target: a direct high-cost edge and an indirect multi-edge route with lower total cost. The algorithm correctly selected the lower-cost path, confirming proper relaxation and cumulative weight handling.

The greedy nearest-neighbor heuristic was tested on a small 4-refugee example. Starting from a fixed node, the algorithm repeatedly selected the closest unvisited refugee based on shortest-path distance. The resulting order matched the expected greedy choice at each step, confirming correct local decision-making.

The 2-opt optimization was verified using a toy route where a known improvement exists. For example, a route A → B → C → D was compared against a reversed segment A → C → B → D, which reduced total cost. The algorithm successfully identified and applied this improvement, confirming correct local search behavior and termination when no better swaps exist.

End-to-end verification was performed on a small evacuation scenario including 1 start node, 3 refugees, and 1 camp. The system successfully generated a valid route that visited all refugees exactly once, avoided blocked edges, and reached a reachable camp, confirming correct integration of all components.

6. Execution Results & Analysis

The system was evaluated using synthetic evacuation maps of increasing size (75, 150, and 250 nodes). In all cases, the algorithm successfully generated valid evacuation routes and adapted to changes in road conditions.

Visualization results show clear differences in routing behavior as road damage increases. Heavily damaged or blocked edges are avoided when possible, while safer routes are prioritized depending on the selected speed-safety parameter *alpha*.

The heuristic optimization approach scales effectively to medium and large graphs, producing high-quality solutions within reasonable computation time. However, as the number of refugee nodes increases, the combinatorial complexity of the routing problem becomes more pronounced.

Overall, the system demonstrates stable performance and meaningful decision-making behavior under varying constraints.

7. Discussion and Conclusions

This project demonstrates the application of graph algorithms and heuristic optimization techniques to a realistic evacuation planning problem. It successfully integrates shortest path computation, combinatorial routing, and multi-objective cost modeling into a unified system.

The results show that heuristic methods provide a practical alternative to exact optimization approaches, particularly in larger problem instances where computational efficiency is required.

Key limitations include the simplified representation of geography and the use of heuristic rather than globally optimal routing. The project primarily applies concepts from Dijkstra's shortest path algorithm and greedy search techniques introduced in the course.

8. AI Disclosure

ChatGPT was used as a supportive tool during development for clarification, debugging assistance, and exploration of algorithmic ideas.

Usage of it consisted of asking it for potential solutions about how to accomplish a given idea when stuck and unable to think of how to start solving a problem (making sure to request pseudo code or text only responses instead of direct code), and prevalently throwing any feedback errors received into it to get a summary of what it actually meant and suggestions for how to solve it.

However, the overall design of the system, the implementation decisions, the testing process, and the actual writing of code was done and validated through independent work, specifically making not to have the AI directly generate code instead of just supporting.

9. Report Quality

The project demonstrates a strong application of graph theory, shortest path algorithms, and heuristic optimization methods within a meaningful real-world context. The system is well-structured, interactive, and capable of handling varying input scales.

The integration of dynamic road conditions and multi-objective decision-making enhances the realism of the model. The visualization component further improves interpretability of results.

Overall, the project shows solid technical understanding, appropriate algorithmic complexity, and effective application of course concepts to a humanitarian logistics problem. It draws primarily on Dijkstra's algorithm and greedy search techniques from the course material.

Lastly, as for the quality of the report I think it's acceptable. It's obviously not 10/10 superb exceeds expectations levels, but I don't really expect it to be given that I'm not particularly good with reports as I tend to gravitate towards short sentences and bullet points to summarize things.